



Temperature Effects on the Molecular Dynamics of Modified Nafion[®] Membranes Incorporating Ionic Liquids' Cations: A ¹H NMRD Study

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Abstract

Nafion[®] is the polymer electrolyte membrane most used in fuel cells (PEMFC), however Nafion[®] presents some limitations due to the water loss with increasing temperature. In this work is presented the study of the molecular dynamics in modified Nafion[®]/ionic liquids (IL) cations membranes with increasing temperature, by proton NMR relaxometry (¹H NMRD). Three Nafion[®] membranes modified with Phenyltrimethylammonium (TPMA⁺), 1-*n*-butyl-3-methylimidazolium (BMIM⁺) and *n*-dodecyltrimethylammonium (DTA⁺) IL cations were considered. This study allowed the evaluation of the effect of the IL cations incorporation in Nafion[®] membranes and to assess the degree of confinement of the IL cations in the membranes matrix in relation with the water content. Thermogravimetry analysis was also performed to study the water loss with increasing temperature

of the Nafion[®]/IL cations membranes. It was possible to establish a correlation between the water content and the IL cations self-diffusion coefficients. The sequential order of the hydration level obtained for the studied systems was Nafion[®]/DTA⁺ > Nafion[®]/BMIM⁺ ≥ Nafion[®]/H⁺ > Nafion[®]/TPMA⁺, while the water loss follows the sequence Nafion[®]/H⁺ > Nafion[®]/BMIM⁺ ~ Nafion[®]/DTA⁺ > Nafion[®]/TPMA⁺. The Nafion[®]/BMIM⁺ presented the largest temperature variation of both the self-diffusion coefficient and the hydration conditions. In terms of PEMFCs efficiency, the Nafion[®]/DTA⁺ modified membrane seems to offer the largest and stable hydration conditions with temperature.

Keywords: ¹H NMRD, Ionic Liquids, Molecular Dynamics, Nafion[®], Thermogravimetry.

1 Introduction

Polymer electrolyte membrane (PEM) fuel cells (PEMFCs) are a promising technology for use as a source of heat and electricity and are a prime candidate to replace the alkaline fuel cell technology, since both convert the energy produced by a chemical reaction into usable electric power [1]. The reference material as PEM (or proton exchange membrane) in fuel cells is Nafion[®], a DuPont[®] copolymer obtained by copolymerization between a perfluorinated vinyl ether comonomer and tetrafluoroethylene (TFE) monomer [2–4]. The chemical structure of Nafion[®] is shown in Figure 1.

The polytetrafluoroethylene (PTFE) backbone has a strong hydrophobic character while the sulfonic acid side-groups

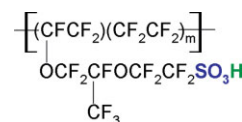


Fig. 1 Chemical structure of Nafion[®] [3].

possess a hydrophilic character. These two opposite properties promote the aggregation of the sulfonic groups into clusters, which contain some water, providing conductive paths for proton migration [5–9], thus conferring high ionic conductivity. These features combined with other properties, such as the mechanical, chemical and thermal stability at tempera-

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tures up to 80 °C, turns Nafion® in the most widely used PEM in PEMFC's.

However, it has been also recognized that Nafion® membranes possess several limitations for their use in fuel cells. The most severe limitation results from the decrease of water content of Nafion® membranes with increasing temperature and, as a consequence, the ionic conductivity is reduced. Therefore, Nafion® membranes can only operate efficiently at temperatures below ~80 °C [7, 8]. There have been many efforts to overcome this problem, where the modification of Nafion® membranes, namely the modification through incorporation of room temperature ionic liquids (IL) is a good example [10–13].

The modification of the Nafion® membranes by incorporation of ionic liquid cations (IL⁺), occurs due to ionic exchange between the hydrogen protons (H⁺) of Nafion® and the IL cation. It is assumed that the IL cations incorporated into the Nafion® matrix replaces hydrogen protons, as it is illustrated in Figure 2. This figure represents the Nafion® structure according to the Yeager three-phase model [14]. The first illustration (a) represents Nafion® in its protonated form showing the presence of hydrogen protons, while in (b), it is represented the modified Nafion®, which is illustrated by the presence of IL cations, larger than hydrogen protons.

The modified Nafion®/IL membranes might be tested and characterized by standard experimental techniques (e. g. conductivity, thermogravimetry). However, these techniques do not provide information about the behavior of the IL cations in the polymer electrolytes. In order to contribute to the understanding of the influence of each IL cation in the modified Nafion®/IL membranes, it is required to use an experimental technique that is able to provide information at molecular level about the dynamics of the nano-confined molecules in the Nafion® cluster walls. In this sense, NMR is a well-established technique used to study the molecular dynamics in a wide range of systems. Proton nuclear magnetic relaxation dispersion (¹H NMRD) has been successfully used for the study of molecular dynamics of liquid crystals in confined environments and of modified Nafion®/IL cation membranes [15–18].

In ¹H NMRD, the proton spin-lattice relaxation time (T_1) is measured over a wide range of Larmor frequencies, typically between 10 kHz and 300 MHz, varying the external magnetic field [19, 20]. T_1 depends on the dipolar interaction between the nuclear spins of the hydrogen atoms. This interaction may be intra- or intermolecular and is affected by the molecular motions. The advantage of ¹H NMRD studies is to be able to detect molecular motions with different characteristic times (typically 10⁻¹¹–10⁻⁴s).

It is important to note that Nafion® membranes constitute complex environments where the clusters contain different molecular species, e.g. hydrogen ions (H⁺) and water molecules. In the case of Nafion® membranes modified with IL cations, additional hydrogen atoms of the IL cations can also be found. However, due to the fact that the proton relaxation times of the small molecules containing hydrogen (H⁺ and

H₂O molecules) are negligible when compared with the proton relaxation times of the larger IL cations, in the Nafion® membranes electrolyte, the ¹H NMRD studies provide information mainly about the molecular dynamics of the IL cations [18]. In previous works, the water dynamics in Nafion® membranes was studied using proton nuclear magnetic relaxation dispersion (¹H NMRD) [21]. Later, it was observed that the incorporation of IL cations in Nafion® membranes might contribute to water retention at high temperatures, depending on the type and incorporation degree of the IL cation [11, 18, 22]. The cations (e.g. H⁺ and IL⁺) mobility in the studied membranes was also characterized by ¹H NMRD at room temperature, and allowed for an estimation of the proton self-diffusion coefficient [18]. However, in order to understand the influence of IL cations on ion mobility in Nafion® membranes with increasing temperature, a detailed molecular dynamics temperature study is required.

Aiming to obtain a deep insight on the IL cations molecular dynamics and its relation with the water content of the modified Nafion®/IL cations membranes ¹H NMRD measurements were carried out. The Nafion® membrane was modified by incorporation with three IL cations: Phenyltri-

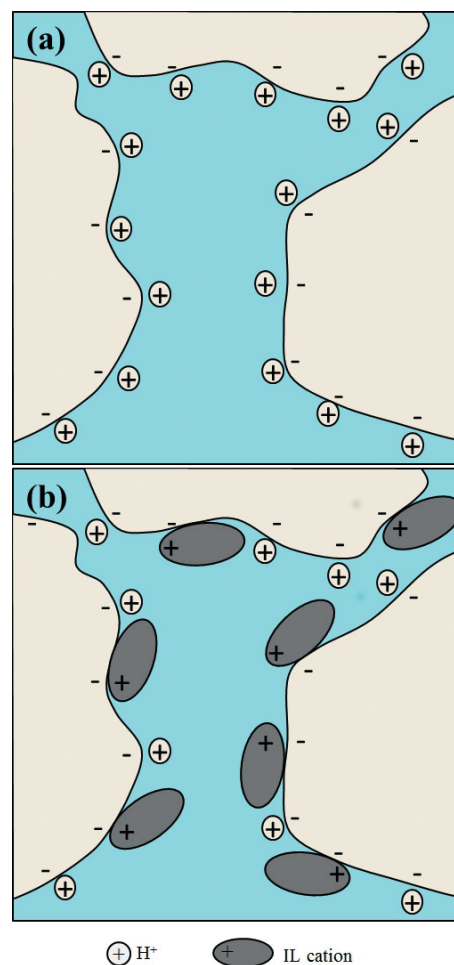


Fig. 2 Schematic representation of IL cation incorporation into the Nafion® matrix: (a) Nafion® in its protonated form; (b) Modified Nafion® incorporated with IL cation.

methyllumonium (TMPA⁺), 1-*n*-butyl-3-methylimidazolium (BMIM⁺) and *n*-dodecyltrimethylammonium (DTA⁺). These IL cations have different characteristics, in particular their hydrophilic character and were selected in order to evaluate the effect of the cations molecular structure in Nafion® membranes.

In this work, the ¹H NMR spin-lattice relaxation rate (1/*T*₁) was measured as a function of frequency for different temperatures in the range 30–80 °C in order to characterize the temperature behavior of the modified Nafion®/IL cations membranes in terms of the IL cations mobility. The ¹H NMRD results were analyzed together with the thermogravimetry analysis (TGA), performed in the same temperature range. TGA provides information exclusively related to the decrease of water content.

2 Experimental

2.1 Materials

Nafion® membranes (N112 and NR212) produced by DuPont (USA) were used. Both types of Nafion® membranes are in their protonated form and have ion exchange capacities (IEC) of 0.92 mmol g^{−1} (i.e. 0.92 mmol of H⁺ per gram of polymer), 51 μm of thickness and an equivalent weight of 1100 g mol^{−1} [23].

Nafion® membranes were modified with three different IL cations. The pure compounds used as source material are: Phenyltrimethylammonium Chloride (TMPA⁺Cl[−]), from Fluka (Germany); 1-*n*-butyl-3-methylimidazolium Tetrafluoroborate (BMIM⁺BF₄[−]), from IoLiTec (Germany); *n*-dodecyltrimethylammonium Chloride (DTA⁺Cl[−]), from TCI Europe (England). TMPA⁺Cl[−] and DTA⁺Cl[−] are two water soluble salts whose cations are quaternary amines. TMPA⁺ cation has a benzene ring and it is the smallest cation. DTA⁺ possesses a hydrocarbon chain which has twelve carbons making it the largest cation of the three types of cation in this study. BMIM⁺

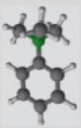
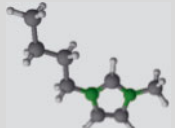
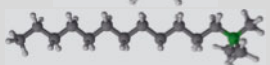
is an IL cation which has an imidazolium ring. Molecular formula, chemical structure, molecular weight length and molar volume of the IL cations are presented in Table 1.

2.2 Membrane Preparation

Prior to the incorporation of the DTA⁺ and TMPA⁺ cations in Nafion® membranes, it was necessary to prepare aqueous solutions with deionized water (40% w/w) of the starting salts, DTA⁺Cl[−] and TMPA⁺Cl[−]. In the case of BMIM⁺ BF₄[−] this procedure was not required as it is an ionic liquid at room temperature.

The modified Nafion® membranes were prepared according to the following procedure. Each Nafion® membrane was immersed in 30 mL of each IL solution during a time no longer than 35 h, to promote the exchange between the free H⁺ from Nafion® and the IL cations from the solution. Different degrees of incorporation of the three IL cations in the Nafion® membranes were expected as a function of contact time between Nafion® membranes and the IL cations solution. The Nafion®/IL cations incorporation degree, defined as the percentage of H⁺ in Nafion® exchanged by IL cations with respect to the maximum IEC was followed through conductivity measurements of the solution which contains the IL cations and the membrane, using a Schott Instruments conductivity meter (model 960). Considering the mass balance and knowing the molar conductance values of each specie (proton and cation) and the solution conductivity throughout time, it was possible to determine the protons quantity that exchange with the cations of the IL in solution [11,24]. Once knowing the exchanged protons quantity, it was possible to determine the incorporation degree of the cations assuming that Nafion® membranes possess a maximum incorporation degree of 0.92 mmol g^{−1}. The incorporation degrees obtained for the systems NR212/BMIM⁺, N112/DTA⁺, and N112/TMPA⁺ were 9, 68, and 82%, respectively.

Table 1 Characteristics of the IL cations studied.

IL cation	Molecular formula	Molecular structure ^(a)	Molecular weight (g mol ^{−1})	Molecular length ^(b) (Å)	Molar volume ^(c) (cm ³ mol ^{−1})
Phenyltrimethylammonium, TMPA ⁺	(CH ₃) ₃ NC ₆ H ₅ ⁺		136	7.2	182
1- <i>n</i> -Butyl-3-methylimidazolium, BMIM ⁺	C ₈ H ₁₅ N ₂ ⁺		139	10.2	189
<i>n</i> -Dodecyltrimethylammonium, DTA ⁺	C ₁₂ H ₂₅ N(CH ₃) ₃ ⁺		228	18.2	350

^(a)Structures were drawn using ChemSketch software;

^(b)Estimated by using the Avogadro software;

^(c)Estimated by Schroeder method.

2.3 NMR Measurements

In this work, we will focus on the ^1H spin-lattice relaxation time (T_1), which was measured for different Larmor frequencies (ν_L). Since $\nu_L = \gamma B/2\pi$ where γ is the proton gyromagnetic ratio and B is the external magnetic field, in order to be possible to measure the proton T_1 in the frequency range 10 kHz to 300 MHz, it is required to vary the magnetic field from ~ 0.2 mT to ~ 7 T. To this end three different NMR spectrometers were used. Between 10 kHz and 9 MHz the T_1 measurements were performed using a home-developed fast-field cycling spectrometer [25]. For frequencies between 10 and 100 MHz the T_1 measurements were performed using a Bruker Avance II console with a 0–2.1 T Bruker BE30 electromagnet. The measurements at 300 MHz were made using a 7 T superconductor magnet equipped with a Bruker Avance II console. In the last two spectrometers, the inversion-recovery pulse-sequence with phase cycling for suppression of the dc bias was used.

The T_1 measurements were performed in a range of temperatures between 30 and 80 °C. The T_1 measured over a four decades of Larmor frequencies makes possible to obtain more detailed information regarding the molecular dynamics in the system.

2.4 Water Content Measurements

The water content of the ionic liquid $\text{BMIM}^+\text{BF}_4^-$ was measured with a Karl-Fisher Metrohm® 831KF Coulometer. A sample of this IL was submitted under vacuum for a period of 48 h for dehydration. The water content was measured in two samples, dehydrated and not dehydrated.

The mass weight loss of Nafion®/IL cations and Nafion®/ H^+ membranes as a function of temperature was determined by thermogravimetry. This analysis was performed in a range of temperatures between 30 and 150 °C, using a Shimadzu TGA-50 system, with a nitrogen atmosphere (flow rate 20 mL min^{-1}) and a heating rate of 5 °C min^{-1} .

3 Results and Discussion

3.1 ^1H NMRD Experiments

Regarding the ^1H NMRD experiments, the results obtained are shown in Figures 3 and 4. The spin-lattice relaxation rates, as a function of frequency (ν_L), in the range 10 kHz–90 MHz, are presented for different temperatures between 30 and 80 °C, for the modified NR212/BMIM $^+$, N112/DTA $^+$, and N112/TMPA $^+$ membranes. It is clear that for the three systems, R_1 exhibits strong frequency dependence. This strong frequency dependence is essentially attributed to the molecular dynamics of the IL cations in the membrane, since the spin-lattice relaxation rate of the small H_2O and H^+ ions can be considered negligible for these systems [18]. However, significant differences can be observed in the ^1H NMRD results between the three systems.

The observed variation of the spin-lattice relaxation rate, for the three modified Nafion®/IL cation membranes with increasing frequency, indicates that at low frequencies the R_1 values correspond to motions with large correlation times. In other words, at low frequencies the R_1 results are associated with the slow motions of the IL cations. At high frequencies the R_1 values are shorter and correspond to faster molecular movements, with shorter correlation times.

The effect of confinement is clearly observed when comparing the R_1 profiles for the NR212/BMIM $^+$ and BMIM $^+\text{BF}_4^-$ (see Figure 5). Not only the shape of the profiles are different but the values of R_1 obtained for the confined IL cations are systematic larger, in particular at low Larmor frequencies.

Regarding the R_1 behavior with increasing temperature, the results of the NR212/BMIM $^+$ system evidence a considerable temperature dependence, in particular in the low frequency range where a factor of ~ 5 is observed between the R_1 at 30 °C and R_1 at 80 °C. $R_1(\nu_L)$ decreases with increasing temperature in the whole frequency range.

The spin-lattice relaxation rate in the N112/DTA $^+$ and N112/TMPA $^+$ systems presents a considerably smaller temperature dependence in comparison with that of NR212/BMIM $^+$.

In the case of the N112/DTA $^+$ system, $R_1(\nu_L)$ presents a smaller temperature dependence in comparison with that of N112/TMPA $^+$. Also, in this system, $R_1(\nu_L)$ decreases with increasing temperature in the whole frequency range. In contrast, for the N112/TMPA $^+$ system, it is observed that in the high frequency range, $R_1(\nu_L)$ decreases with increasing temperature and in the low frequency range, $R_1(\nu_L)$ presents the opposite temperature dependency behavior.

The different $R_1(\nu_L)$ profiles obtained at different temperatures reflect the confinement level of the electrolyte cations and water molecules in the Nafion® matrix. In view of the observed variation of the spin-lattice relaxation rate with frequency and temperature of NR212/BMIM $^+$, it seems that this system possesses a lower level of confinement in comparison to the other two modified Nafion®/IL cation membranes. In other words, BMIM $^+$ cations, water molecules and free hydrogen ions motions are less constrained and/or structured than in the case of the other cations within the Nafion® membranes.

In view of the strong temperature effect on the NR212/BMIM $^+$ $R_1(\nu_L)$ profiles and since BMIM $^+\text{BF}_4^-$ is known to be a hydrophilic ionic liquid, $R_1(\nu_L)$ measurements in two samples of pure BMIM $^+\text{BF}_4^-$ with different water content were performed, in order to evaluate the effect of water content in the IL molecular dynamics. One sample presented a water content of 840 ppm and was in the same hydration conditions as the IL used in the incorporation studies. The other was partially dehydrated and presented a water content of only 115 ppm.

In Figure 4 are presented the spin-lattice relaxation rate results for the two samples of BMIM $^+\text{BF}_4^-$ at three different temperatures (30, 50, and 80 °C). Comparing the $R_1(\nu_L)$ profiles, it is clear that the presence of water in the IL does not affect the ^1H relaxation, because the $R_1(\nu_L)$ profiles of the hydrated and dehydrated samples identical, for each temperature, within the experimental uncertainty. In view of these

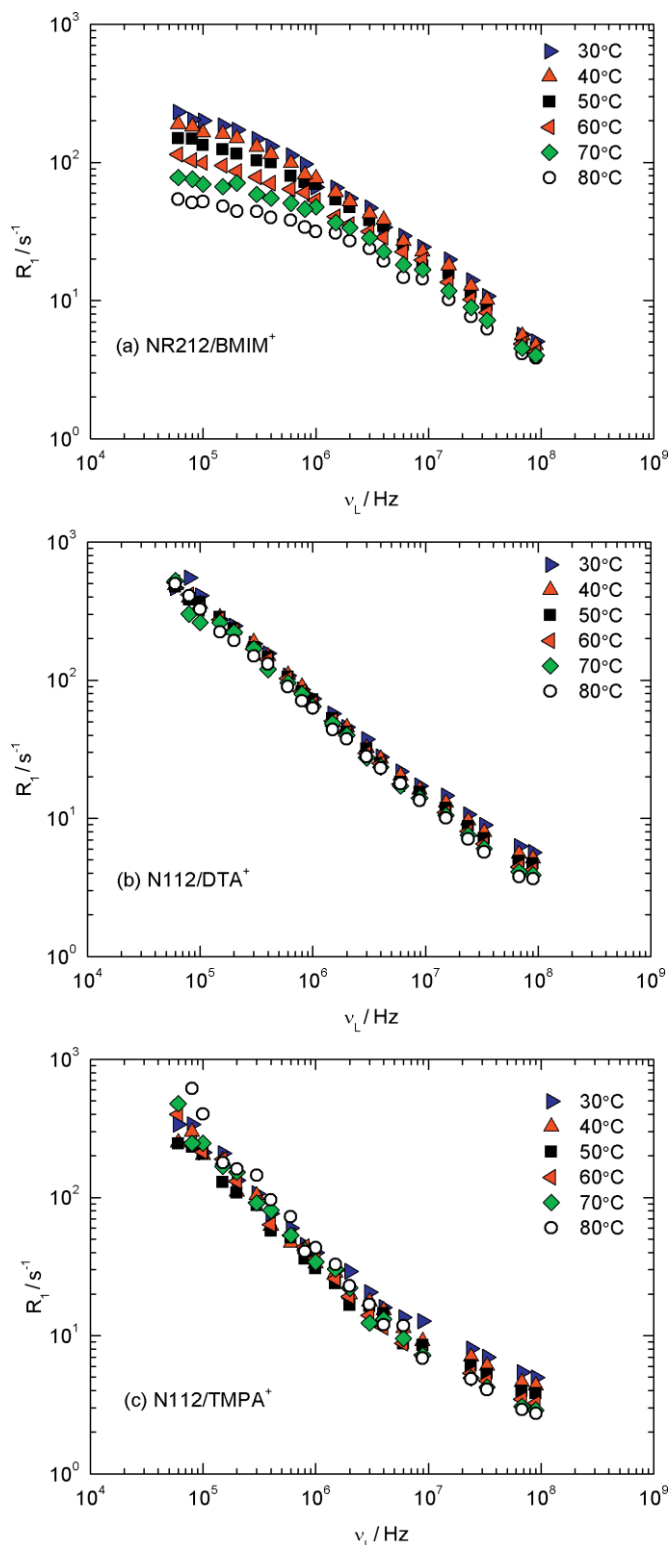


Fig. 3 Spin-lattice relaxation rate (R_1) as a function of frequency for different temperatures in the three modified Nafion®/IL cation membranes.

observations it is possible to infer that the strong temperature dependence observed for the $R_1(\nu_L)$ of the modified NR212/BMIM⁺ system (Figure 3a), is not related with the water content of the IL prior to the incorporation process.

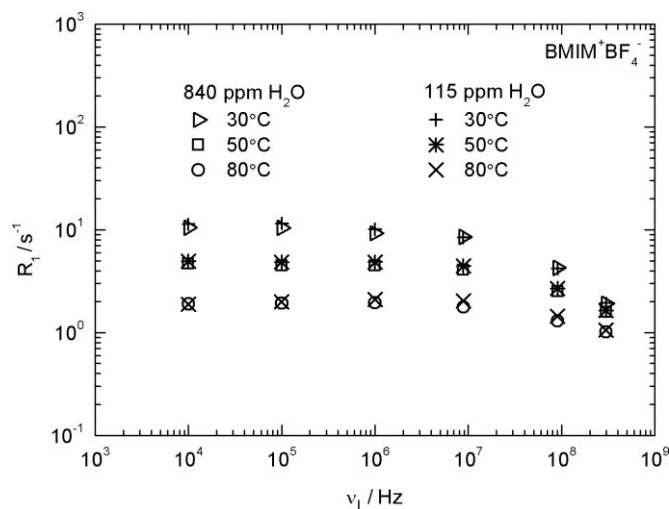


Fig. 4 Spin-lattice relaxation rate as a function of frequency, for two samples of pure BMIM⁺BF₄⁻ with two different water contents (840 and 115 ppm), at different temperatures.

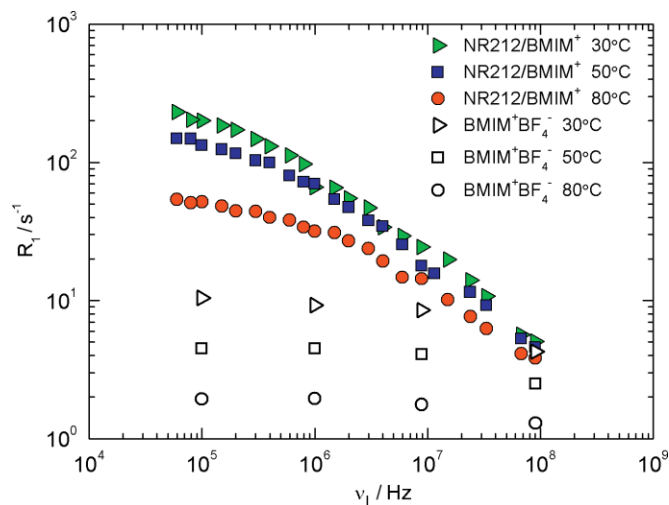


Fig. 5 Spin-lattice relaxation rates as a function of frequency for different temperatures for NR212/BMIM⁺ and IL BMIM⁺BF₄⁻ systems.

In Figure 5 are presented, for comparison, the $R_1(\nu_L)$ results obtained for the hydrated IL BMIM⁺/BF₄⁻ and for the NR212/BMIM⁺ membrane, for the temperatures 30, 50, and 80 °C. The spin-lattice relaxation rates obtained for the NR212/BMIM⁺ membrane are larger than those measured for BMIM⁺BF₄⁻, for all temperatures and frequencies considered. This fact is clearly related to the confinement state of the IL cations, free protons and water molecules in Nafion® membranes, since the mobility inside the membrane is limited by the polymeric clusters' walls, in comparison with the molecular motions in liquid phase.

In the pure IL, $R_1(\nu_L)$ is almost frequency independent for frequencies up to 9 MHz, and slightly decreases for frequencies above 10 MHz. The R_1 obtained for NR212/BMIM⁺ presents a completely different frequency dependence, with the spin-lattice relaxation rate increasing continuously with the

decrease of frequency. For both IL systems, $R_1(\nu_L)$ decreases with increasing temperature. The effect of temperature on the molecular dynamics seems to be more important in the case of NR212/BMIM⁺ since the $R_1(\nu_L)$ profile also changes with temperature. It is interesting to note also that $R_1(\nu_L)$ for NR212/BMIM⁺ system at 80 °C, reveals a smaller frequency dependence below 10 MHz.

3.2 Modeling of the Relaxation Rate of the Modified Nafion/IL Cation Membranes

The spin-lattice relaxation rate can be analyzed in terms of relaxation models associated with the different types of molecular motions in the systems [15–17]. Usually, it is possible to write a global model that takes into account the sum of all contributions to the relaxation rate. In view of the intrinsic complex structure of the Nafion® membranes and of the complex medium formed by the IL electrolytes incorporated in the Nafion® membranes it is necessary to consider not only the molecular motions found in bulk fluids (e.g. molecular rotations/reorientations (R), translational self-diffusion (SD)) but also molecular motions that are strongly affected by the confining environment. Since Nafion® is characterized by a network of strong rigid regions formed by the polytetrafluoroethylene backbone and nano-size cavities interconnected by narrow channels promoted by the micro-segregated sulfonic acid side-groups, where a liquid electrolyte can be found, proton NMR is not directly sensitive to the PTFE matrix but only to the liquid electrolyte molecules. The motions of the electrolyte molecules in a thin layer close to the cluster walls can be very different from the motions in bulk. Close to the walls, the molecular motions are affected by the topology of the cluster walls and channels. In fact, the translational displacement jump of one molecule between two points close to the wall might induce the reorientation of that molecule. The diffusion process close to wall might be also affected by the interactions between the electrolyte ions and the PTFE sulfonic acid side-groups on the Nafion® membranes' cluster walls. These restricted motions contribute to the spin-lattice relaxation with a mechanism that is usually referred as "rotations mediated by translational displacements" (RMTD) [15, 16, 18].

Therefore, the relaxation model can be expressed by a linear combination of R+SD and RMTD relaxation mechanisms:

$$R_1 = (R_1)^{R+SD} + (R_1)^{RMTD} \quad (1)$$

As considered on a previous Nafion®/IL NMR study [18], $(R_1)^{R+SD} = (R_1)^R + (R_1)^{SD}$ can be described as a first approximation by the Bloembergen, Purcell, and Pound (BPP) model [26]

$$(R_1)^{R+SD} = \frac{9}{8} \left(\frac{\mu_0}{4\pi} \right)^2 \frac{\gamma^4 \hbar}{r_{eff}^6} \left(\frac{\tau_c}{1 + 4\pi^2 \nu^2 \tau_c^2} + \frac{4\tau_c}{1 + 16\pi^2 \nu^2 \tau_c^2} \right) \quad (2)$$

where r_{eff} is an effective distance associated with an average inter-proton distance in the molecules and τ_c is a thermally activated effective correlation time with an activation energy

E_R , $\tau_c = \tau_{333K} e^{(E_R/RT)(1/T-1/333)}$, associated with characteristic time of local rotations and translational self-diffusion.

The $(R_1)^{RMTD}$ contribution can be described by the following expression,

$$(R_1)^{RMTD} = A_{RMTD} \left\{ \frac{1}{(2\pi\nu)^p} \int_{z_{min}}^{z_{max}} \frac{z^{3-2p}}{1+z^4} dz + \frac{1}{(4\pi\nu)^p} \int_{\frac{z_{min}}{\sqrt{2}}}^{\frac{z_{max}}{\sqrt{2}}} \frac{z^{3-2p}}{1+z^4} dz \right\} \quad (3)$$

where $z_{max}^2 = v_{c max}/v$ and $z_{min}^2 = v_{c min}/v$. A_{RMTD} is a parameter that depends on the residual dipole-dipole proton spin interaction averaged by the local molecular reorientations, on the microstructural features of the confined environment, and can be expressed by $A_{RMTD} = A_0 D^{-1/2}$, where A_0 is assumed temperature independent, as a first approximation [16], and D is the self-diffusion constant. The exponent p is associated with the orientational structure factor associated with the diffusion/reorientation of the molecules at the surfaces of the cluster walls [15,16,18]. The high and low cut-off frequencies, $\nu_{c min}$ and $\nu_{c max}$, respectively, are related to the self-diffusion constant through the following relations: $\nu_{c max}^{-1} = 2\pi l_{min}^2/(4D)$ and $\nu_{c min}^{-1} = 2\pi l_{max}^2/(4D)$, where l_{min} and l_{max} are respectively the smallest and largest displacement distances. The diffusion coefficient is temperature dependent and it is usually described by an Arrhenius Law

$$D = D_0 e^{-E_D/RT} = D_{333K} e^{-(E_D/R)(1/T-1/333)} \quad (4)$$

where E_D is the activation energy, D_0 is a pre-exponential factor, and $D_{333 K}$ is the value of self-diffusion coefficient at the arbitrary reference temperature of 333 K.

For each modified Nafion®/IL membrane system the relaxation model given by Eq. (1) was fitted to the experimental spin-lattice relaxation results obtained in the temperature range 30–80 °C, taking into account the frequency and temperature dependences simultaneously. For instance, in the case of NR212/BMIM⁺, all 156 R_1 experimental data were considered in the model fit to obtain the nine parameters in Eq. (1). As it was previously reported for other systems [17–20, 27], a non-linear least-square minimization procedure with a global minimum target was used. Identical procedure was used for N112/DTA⁺ and N112/TMPA⁺ systems.

It was observed that the exponent p in the RMTD model was almost temperature independent for all systems taking into account the fitting parameters error estimates. In the case of the N112/TMPA⁺ system the RMTD pre-factor A_0 was found to be slightly temperature dependent. As a first approximation, a second degree polynomial was used.

In Table 2 are presented the values of the model fitting parameters for the three systems. In Figure 6 are presented the best fitting curves for the three modified Nafion®/IL cation membranes at 60 °C.

It is clear from Figure 6 that the relaxation model of Eq. (1) describes very well the R_1 frequency dependence for the three modified Nafion®/IL systems. In the case of NR212/BMIM⁺ it is possible to observe the low frequency plateau associated with the low frequency cut-off of the $(R_1)^{RMTD}$ relaxation mechanism (see Figure 6a).

The values of the self-diffusion coefficient as a function of temperature, obtained from the analysis of the experimental spin-lattice relaxation rate results, for the three modified Nafion® membranes are presented in Figure 7 for the temperature range 30–80 °C. They were calculated using Eq. (4) and the values of $D_{333\text{ K}}$ and E_D presented in Table 2 for each system. It is observed that the self-diffusion coefficient of N112/DTA⁺ and N112/TMPA⁺ cations present a small temperature dependence expressed by the low activation energies obtained from the model fit. In the case of NR212/BMIM⁺, the activation energy is larger by a factor of ~5 in comparison with the value of the activation energy obtained for N112/DTA⁺ and more than a factor of ~20 in comparison with that obtained for N112/TMPA⁺. The value of E_D obtained for the NR212/BMIM⁺ is compatible with the one obtained for the self-diffusion coefficient of the BMIM⁺ cations in the pure BMIM⁺BF₄[−] [28].

The self-diffusion values obtained for the N112/TMPA⁺ membrane is almost independent of temperature as observed in Figure 7. This may be due to the fact that the diffusion of the TMPA⁺ cations is strongly dependent on the Nafion® cluster environment and less dependent on the thermal energy. This might also be due to the reduced water content in the N112/TMPA⁺ electrolyte and reduced water loss with increasing temperature as explained below (Section 3.3). Water molecules, TMPA⁺ cations and free H⁺ are probably more structured and correlated to the cluster walls. This is also evidenced by the low values of the self-diffusion constant obtained for N112/TMPA⁺ in comparison to the other systems.

Regarding the N112/DTA⁺ system, the estimated self-diffusion constant is about one order of magnitude larger than the one estimated for the N112/TMPA⁺ system. This fact was expected since the N112/DTA⁺ system presents a water content larger than the water content of the other two systems studied, as it will be discussed in the following. Since, it is reasonable to assume that the viscosity of the electrolyte decreases with an increase of water content, self-diffusion should increase, in view of the Stokes-Einstein relation. Concerning the estimated N112/DTA⁺ self-diffusion activation energy, this parameter is only slightly larger than that estimated for N112/TMPA⁺ which also reveals a weak diffusion's temperature dependence. As in the case of N112/

TMPA⁺ the small activation energy might be related with the degree of the structural correlation of the TMPA⁺ cations with the cluster walls. Nevertheless, the difference between the activation energies of both systems might be related with the difference between the water contents in the two systems.

Concerning the NR212/BMIM⁺ system, the estimated self-diffusion obtained at room-temperature is similar to the one obtained for the N112/DTA⁺ system. However, it presents a much larger temperature dependence expressed by a significantly larger activation energy in comparison with that of N112/DTA⁺. This fact, and the observation that the activation energy in NR212/BMIM⁺ is close to the one determined for the pure BMIM⁺BF₄[−] [28], might indicate that the self-diffusion of the BMIM⁺ cations in the NR212/BMIM⁺ membrane is controlled mainly by thermal effects and less affected by the interaction with the membrane cluster walls, in contrast with N112/DTA⁺ and N112/TMPA⁺ modified membranes.

It is interesting to observe that the value of self-diffusion estimated for the NR212/BMIM⁺ membrane at 30 °C is one order of magnitude larger than the one obtained by Tokuda et al. [28] for BMIM⁺ cations in bulk. However, in that work it is shown that the value of BMIM⁺ self-diffusion depends on the nature and mobility of the IL anions. Therefore, in view of the structure of the Nafion®'s polymer and absence of mobility of sulfonic acid side-groups on the surface of the clusters' walls, it is reasonable to expect a difference between the BMIM⁺'s diffusion in bulk and within the membrane. In addition, the solvation degree in the modified NR212/BMIM⁺ membrane is larger than in the pure BMIM⁺BF₄[−] due to the higher water molar fraction with the respect to the BMIM⁺ cations in the Nafion® membrane.

The Bloembergen, Purcell and Pound (BPP) relaxation model used to describe the contributions of bulk-like rotations and self-diffusion allows for the estimate of correlation times for these motions in the range of 2×10^{-10} – 1.6×10^{-9} s with activation energies between 1.4 and 30 kJ mol^{−1}. These values are compatible with those obtained for small 5CB liquid crystal molecules either in bulk or under confinement in nano-porous glasses, taking into account that the IL cations in the Nafion®/IL systems are smaller than the 5CB molecules [16, 29]. The BPP pre-factor allows for the estimation of an effective inter-proton distance in the range of 2.7–2.9 Å, for the three Nafion®/IL systems.

Table 2 Parameters obtained by fitting the spin-lattice relaxation model (Eq. 1) to the experimental results.

	NR212/BMIM ⁺	N112/DTA ⁺	N112/TMPA ⁺
p_0	0.50 ± 0.02	0.86 ± 0.01	0.79 ± 0.02
$r_{\text{eff}} (10^{-10} \text{ m})$	~2.9	~2.7	~2.8
$A_0 (\text{m}^{-1} \text{ s}^{-(3/2+p)})$	$0.27 \pm 0.04^{(a)}$	$60.4 \pm 0.8^{(a)}$	$2.3 \pm 0.4^{(a)}$
$t_{333\text{ K}} (10^{-10} \text{ s})$	4.5 ± 0.2	6.7 ± 0.8	7.9 ± 0.1
$E_R (\text{kJ mol}^{-1})$	21.8 ± 0.3	14 ± 11	15 ± 0.2
$I_{\text{max}} (10^{-9} \text{ m})$	31 ± 0.8	31	22
$I_{\text{min}} (10^{-9} \text{ m})$	1.02	1.82	0.72
$D_{333\text{ K}} (10^{-11} \text{ m}^2 \text{ s}^{-1})$	47 ± 2	24 ± 2	1.17 ± 0.04
$E_D (\text{kJ mol}^{-1})$	30 ± 0.7	5.9 ± 0.9	1.4 ± 0.8

^(a) A_0 was found to be temperature independent in the case of NR212/BMIM⁺ and N112/DTA⁺ systems and described by $A_0 = 2.4 + 0.055(T - 333) + 1.38 \times 10^{-3}(T - 333)^2$ for N112/TMPA⁺ system.

3.3 Thermogravimetry Experiments

Thermogravimetry studies were performed to evaluate the effect of incorporation of IL cations in Nafion® membranes on the water loss with increasing temperature.

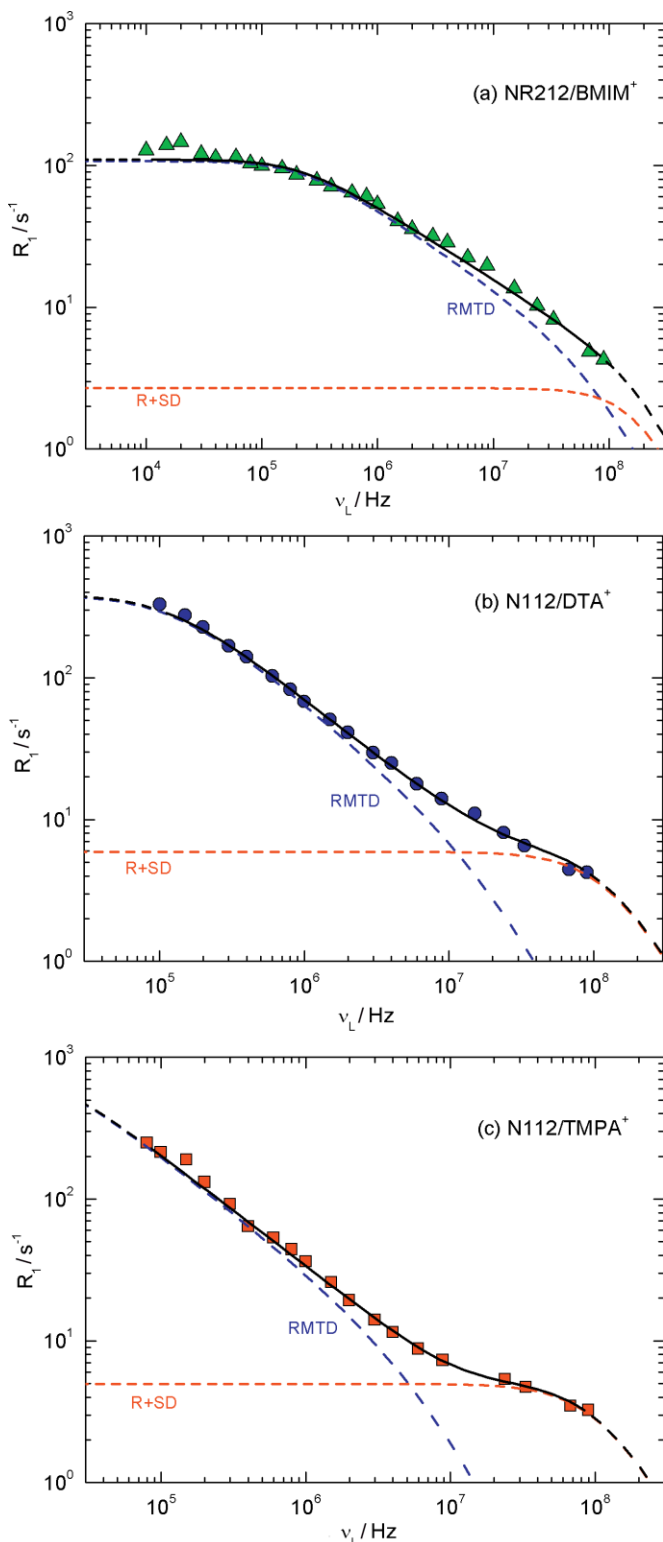


Fig. 6 Experimental results and theoretical fitting curves for the three modified Nafion® membranes at 60 °C.

Figure 8 presents the normalized membranes' weight as a function of temperature for the three modified Nafion® membranes and for the Nafion® membrane in its protonated form. The normalization was done respect to the system's weight at 20 °C. In the temperature range of 30–150 °C, it is assumed that the observed weight decrease is only due to water loss from the membranes, as it is often considered in similar analyses [30–33]. It is clear from the analysis of Figure 8 that there are two extreme cases: the NR212/H⁺ presents the largest water loss and the N112/TMPA⁺ modified membrane presents almost no water loss in the temperature range studied. The N112/DTA⁺ and NR212/BMIM⁺ present water losses in between the two previous cases. A more detailed analysis of the changes of the membranes' weight with increasing temperature in the N112/DTA⁺ and NR212/BMIM⁺ systems shows that the weight of N112/DTA⁺ seems

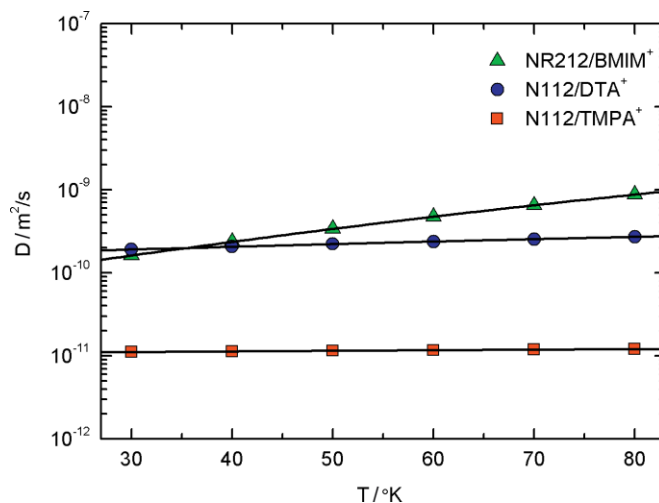


Fig. 7 Simulated values of the self-diffusion coefficients as a function of temperature obtained with Eq. (4) using the activation energies and diffusion coefficients obtained from the model fit (Eq. 1) to the R_1 experimental results in the three Nafion®/IL systems.

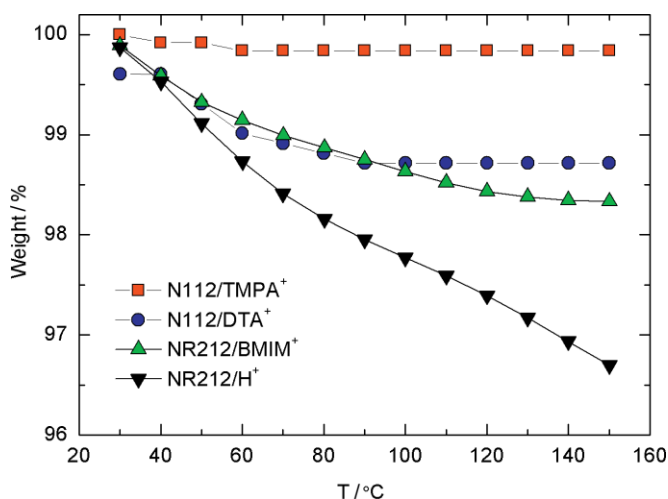


Fig. 8 Membrane's weight as a function of temperature for the studied systems.

to stabilize at high temperatures (above 100 °C) while, in the case of NR212/BMIM⁺ system, appears to achieve a slight stabilization at temperatures above 140 °C. The total water loss of the NR212/BMIM⁺ is slightly larger than that of N112/DTA⁺. The water loss in the three modified Nafion®/IL cations membranes is clearly distinct from that of the NR212/H⁺ system, where it is observed a continuous weight decrease with increasing temperature. For the NR212/H⁺ membrane the weight loss does not stabilize at high temperature, in contrast with the Nafion®/IL cations systems.

In Figure 9, it is presented the ratio between the water weight and weight of dry membrane for the four Nafion® membrane systems: N112/TMPA⁺, N112/DTA⁺, NR212/BMIM⁺, and NR212/H⁺. It is clear that the N112/DTA⁺ membrane presents the largest relative water content of the four systems. In contrast, the N112/TMPA⁺ membrane presents the lowest water content. The NR212/BMIM⁺ membrane presents a slightly larger relative water content in comparison with the NR212/H⁺ membrane and it is interesting to mention that NR212/BMIM⁺ membrane at high temperatures (above 90 °C) presents a relative water content approximated to the residual water content (~5%) of Nafion® in its protonated form, at room temperature. At high temperatures the relative water content in both N112/DTA⁺ and NR212/BMIM⁺ membranes systems seem to stabilize in contrast to the NR212/H⁺ membrane. The N112/TMPA⁺ membrane presents almost no changes in its relative water content.

For the N112/DTA⁺ and NR212/BMIM⁺, the incorporation of the IL cations happened in parallel with the incorporation of water molecules. However, for the N112/TMPA⁺ the incorporation of TMPA⁺ cations seem to correspond to a decrease of the initial NR212/H⁺ residual water content.

From the fact that the water content is almost temperature independent in the case of N112/TMPA⁺ system (as observed in Figure 8), it is possible to assume that the water molecules are more strongly confined and structured. This strong confinement state of the water molecules might also be related with the reduced water content and high incorporation

degree (82%) of the TMPA⁺ cations in the N112 membrane, which are solvated by the confining water molecules.

On the other hand, the thermogravimetry results obtained for the NR212/BMIM⁺ support the idea that the water molecules in this membrane are less confined and less bounded as the result of the solvation equilibrium between the Nafion®-SO₃⁻/BMIM⁺ ions and the water molecules.

Overall, the thermogravimetry results show that the incorporation of the IL cations promotes an increase of the confinement of the water molecules and an increase of their bounded state in the membrane's electrolyte due to the Nafion®-SO₃⁻/IL ions solvation conditions within the matrix of Nafion®, thus contributing to the stabilization of the water content at high temperatures. The solvation process involving the electrochemical dissociation of anion/cation pairs by the water molecules and the molecular correlation equilibrium reached for each system contributes to define the ability of each modified membrane to retain water inside the its matrix.

The results obtained by thermogravimetry are consistent with the results obtained by ¹H NMRD in what concerns the self-diffusion coefficient estimated for the three modified Nafion®/IL cations' membranes.

In fact, regarding the N112/TMPA⁺ membrane, the small self-diffusion coefficient estimated from the analysis of the ¹H NMRD results, in comparison with the values estimated for the other two Nafion®/IL cations systems and the very small activation energy estimated from the temperature dependence study of the ¹H spin-lattice relaxation rate allow for the same conclusions regarding the confinement state and bounding conditions of the membrane's electrolyte. As previously stated, the very small activation energy estimated suggests a limited solvation state of the Nafion®s - SO₃⁻ and TMPA⁺ ions, in agreement with the low water content of the membrane and the small temperature dependence of the water loss.

Regarding the NR212/BMIM⁺ and N112/DTA⁺ membranes, the estimated self-diffusion coefficients are about one order of magnitude larger than that of the N112/TMPA⁺ membrane. This result seems to be correlated with the amount of water incorporated in these membranes. The fact that the NR212/BMIM⁺ system presents about three times less water than the N112/DTA⁺ seems not to be related with the value of the estimated self-diffusion coefficient. This aspect is also supported by the very little influence of the water content in the spin-lattice relaxation measurements performed in the pure BMIM⁺BF₄⁻ with the respect to the temperature independence of the spin-lattice relaxation rate (see Figures 4 and 5). The activation energy associated with the self-diffusion coefficient in the NR212/BMIM⁺ is considerably larger than that estimated for the N112/DTA⁺ (see Table 2). This fact seems to be related mainly with the solvation conditions of BMIM⁺ cations and not with the water content in the NR212/BMIM⁺ which is smaller than in N112/DTA⁺.

The mobility of the BMIM⁺ cations in the Nafion® membrane seems to be larger than that of the other modified

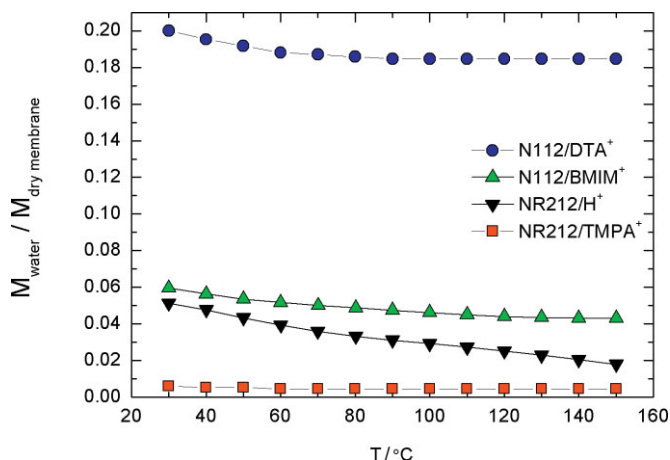


Fig. 9 Evolution of relative water content (w/w) with increasing temperature.

Nafion®/IL cations membranes. This larger mobility seems to correspond to the slightly larger water loss in comparison with the N112/DTA⁺ system.

Regarding to N112/DTA⁺ and N112/TMPA⁺ systems, the fact that the self-diffusion activation energy of N112/DTA⁺ is larger than the one obtained for N112/TMPA⁺ is consistent with the particular temperature dependence of the solvation conditions, taking into account not only the electrostatic interaction between the electrolyte molecules and the walls but also the water content. In fact, N112/DTA⁺ has a larger water content and the water loss is larger than that of N112/TMPA⁺ but the total amount of water is not very much reduced (~1.25%). Therefore, the solvation conditions slightly change with temperature. N112/TMPA⁺ has lower water content and a reduced water loss then the solvation conditions remain almost constant for this system.

4 Conclusion

In this paper are presented results of thermogravimetry and ¹H NMRD studies in modified Nafion® membranes by incorporation of three different IL cations: Phenyltrimethylammonium, 1-*n*-Butyl-3-methylimidazolium, and *n*-Dodecyltrimethylammonium. The aim was to understand the effect of temperature and water content on the molecular structure and dynamics of IL cations in Nafion® membranes. The thermogravimetry studies revealed different degrees of water incorporation depending on the IL cations and different water retention behaviors with temperature.

The ¹H spin-lattice relaxation was measured as a function of temperature and of the Larmor frequency in the range of 10 kHz–300 MHz for the three Nafion®/IL cations systems. From the analysis of the obtained results, it is possible to conclude that the molecular motions of the IL cations in the membranes' clusters, are local molecular rotations/reorientations and rotations mediated by translational displacements influenced by the cluster's walls.

It was possible to establish a correlation between the water content with the estimated self-diffusion coefficients including the observed temperature dependences of the two observables. The reduced value of water content and water loss obtained for the N112/TMPA⁺ system was consistent with the low value of the self-diffusion coefficient and with the low value of activation energy. In this system, both Nafion®'s – SO₃[−] anions and TMPA⁺/H⁺ cations and water molecules seem to form a strongly bounded electrolyte. In the case of NR212/BMIM⁺ and N112/DTA⁺ systems, it was possible to conclude that the water content of these systems, depends on the chemical structure of the IL cations. Also, the water loss with temperature is similar in the two systems, although slightly larger in the case of BMIM⁺. The larger value of water content in the NR212/BMIM⁺ and N112/DTA⁺ systems, in comparison with N112/TMPA⁺, is associated with the larger self-diffusion coefficient obtained for these systems. The larger activation energy estimated for the NR212/BMIM⁺ seems

to be related mainly with the thermal effect on the self-diffusion and less dependent from the interactions between the BMIM⁺ IL cations and the sulfonic groups in the Nafion® cluster walls, in contrast with N112/DTA⁺ and N112/TMPA⁺ modified membranes.

From the PEMFC's applications point of view, the structure of the DTA⁺ cations when incorporated in the Nafion® membranes seems to favor the incorporation of a large amount of water molecules in comparison with the other modified Nafion®/IL cations membranes and also the NR212/H⁺. The solvation conditions of the Nafion®'s – SO₃[−] and DTA⁺ ions appears to contribute to the water retention and to a small temperature dependence of the self-diffusion coefficient.

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